

TECHNICAL REPORT

AEMS Heritage Pipeline: Automated Geospatial Mapping for Cultural Heritage Preservation in Offa, Nigeria

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Principal Investigator

Celio Soares de Souza, MSc Candidate

UNESP — São Paulo State University, Brazil

Graduate Program in Environmental Sciences

Email: celio.s.souza@unesp.br

ORCID: 0009-0000-1565-3803

Local Partner

Salako Lukman Olamilekan

Offa Heritage Documentation Initiative

Wikimedia Community User Group Nigeria

Institutional Affiliations

- São Paulo State University (UNESP), Brazil
- Offa Heritage Documentation Initiative, Nigeria
- Wikimedia Community User Group Nigeria

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1. EXECUTIVE SUMMARY

This report documents the development, deployment, and validation of the AEMS Heritage Pipeline, an automated computational framework for the geospatial mapping of cultural heritage assets in Offa, Kwara State, Nigeria. The pipeline represents a concrete instantiation of the Adaptive Environmental Modeling System (AEMS) framework, adapted for Digital Public Infrastructure (DPI) applications in cultural heritage preservation within data-scarce environments of the Global South.

The pipeline processed 500 community-documented photographs from Wikimedia Commons, automatically extracting metadata, identifying location references through pattern matching, and performing offline geocoding via a specialized local gazetteer. The system produced an interactive, self-contained HTML map of heritage assets, requiring zero dependency on external geocoding APIs after the initial metadata extraction phase.

Key quantitative results:

- Total images processed: 500 (100% of the target category)
- Unique locations identified: 46
- Locations successfully geocoded: 40 (87.0% success rate)
- Images with inferred GPS coordinates: 321 (64.2% of total)
- External API dependency after Phase 1: Zero
- Total execution time: approximately 8 minutes on standard consumer hardware
- Reproducibility: Deterministic given identical input data

The pipeline demonstrates that the core principles of the AEMS framework—adaptivity, uncertainty quantification, and offline-first computation—extend beyond environmental monitoring to Digital Public Infrastructure for cultural sovereignty. By transforming unstructured community photographs into geospatially-enabled, semantically-structured knowledge assets, the system provides a scalable model for community-driven heritage preservation that respects local ownership of cultural narratives.

This technical report is intended to support funding submissions to the Wikimedia Foundation, UNESCO, and related cultural heritage preservation initiatives, and to serve as a reproducible reference for similar deployments in other municipalities of the Global South.

2. CONTEXT AND SCIENTIFIC MOTIVATION

2.1 The Digital Sovereignty Challenge in the Global South

Cultural heritage documentation in rapidly developing regions of the Global South faces a structural paradox: communities are actively producing rich visual documentation of their tangible and intangible heritage, yet this documentation remains semantically unstructured, geographically invisible to global knowledge systems, and disconnected from the formal infrastructures of digital preservation.

In Offa, a historic Yoruba city in Kwara State, Nigeria, the local Wikimedia community has undertaken a systematic documentation effort through photo walks and physical registries, capturing over 500 heritage assets including ancient palaces, historical mosques and churches, traditional shrines, schools, healthcare centers, and cultural landmarks associated with the Ijakadi Festival and historical figures such as the Asiwajus of Offa. However, these images suffer from three critical limitations:

First, the vast majority lack embedded GPS metadata, as cameras and smartphones frequently disable location tracking for privacy reasons, and image sharing via messaging platforms strips EXIF data. Second, the images lack structured semantic links to Wikidata, preventing their integration into global knowledge graphs. Third, they lack geospatial indexing, rendering them invisible to municipal planning systems and regional heritage preservation frameworks.

2.2 The AEMS Framework as a Unifying Architecture

The Adaptive Environmental Modeling System (AEMS) was originally conceived as a computational framework for environmental intelligence in data-scarce settings, operationalizing Norbert Wiener's cybernetic principles through modern sequential Bayesian data assimilation. Its core design principles—probabilistic reconstruction of missing observations, explicit uncertainty quantification, and offline-first computation—have been validated across multiple domains including oceanographic HAB monitoring in Central Patagonia (Quezada, Souza, and Ruiz, 2026), predictive One Health early warning for Sudan ebolavirus in Uganda (Souza and Ibrahim, 2026), and predictive environmental governance for pastoral resilience in Kenya (Souza and Roba, 2026).

The heritage documentation challenge in Offa shares fundamental structural characteristics with these environmental monitoring problems: fragmented observations, missing metadata, the need for probabilistic reconstruction, and the requirement for offline-first computation in low-connectivity environments. This structural isomorphism motivated the adaptation of the AEMS framework to cultural heritage preservation, treating heritage assets as a form of "built environment intelligence" analogous to environmental variables in traditional AEMS deployments.

2.3 From Environmental Intelligence to Cultural Sovereignty

The adaptation of AEMS to heritage documentation represents more than a technical exercise; it embodies a decolonial approach to Digital Public Infrastructure. Traditional heritage preservation frameworks frequently rely on Western-centric archival institutions (museums, academic repositories, proprietary cultural databases) that extract knowledge from communities without

returning structured, accessible infrastructure. The AEMS Heritage Pipeline inverts this dynamic by providing communities in the Global South with the technical tools to control, structure, and globally publish their own historical and structural narratives through open Wikimedia infrastructure.

This approach aligns with emerging paradigms in decolonial data science, which emphasize technical architectures that center Global South knowledge systems rather than extracting from them (Mohamed, Png, and Isaac, 2020), and with the Wikimedia Foundation's strategic priorities around knowledge equity and community-driven content creation.

3. PROBLEM STATEMENT

Transforming localized town histories and physical civic assets into a globally accessible, verifiable digital public good faces two structural and technical bottlenecks that motivate the present work.

3.1 Challenge 1: Manual Scaling Barriers and Knowledge Fragmentation

Documenting an entire municipality across multiple local districts—in the case of Offa, the six districts of Shawo, Essa, Balogun, Ojomu, Asalofa, and Igbodun—involve massive volumes of fragmented assets including schools, basic health centers, hotels, mosques, churches, and traditional shrines. Relying solely on manual content creation and basic photo uploads slows down the population rate of open repositories and struggles to generate the deep semantic links required by global search engines and Wikidata structures.

The manual documentation workflow exhibits three specific bottlenecks: (i) the absence of automated metadata extraction, requiring labor-intensive manual tagging of each image; (ii) the lack of geospatial indexing, preventing integration with municipal planning systems; and (iii) the absence of semantic structuring, preventing automated integration with Wikidata knowledge graphs.

3.2 Challenge 2: The Physical-to-Digital Disconnect and Information Integrity

While placing QR codes on historical frames represents a major innovation for community engagement, maintaining the integrity, verification, and dynamic updates of the linked spatial data requires an underlying analytical infrastructure. Without an automated data processing engine, local governments and cultural institutions cannot evaluate the structural preservation of assets over time, predict degradation risks, or comprehensively map public infrastructure availability.

This physical-to-digital disconnect is compounded by the absence of automated verification mechanisms. QR codes linked to static web pages become broken links when content is updated, moved, or deleted, creating misinformation vectors rather than knowledge infrastructure. The AEMS Heritage Pipeline addresses this through the QR-Linked Dynamic Verification Engine (described in the associated

concept note), which monitors QR code scans and flags broken links or outdated articles.

3.3 Challenge 3: API Dependency and Infrastructure Fragility

A third challenge, not anticipated in the original concept note but revealed during implementation, concerns the fragility of external geocoding APIs in low-resource deployment contexts. Initial attempts to use the Nominatim (OpenStreetMap) and Photon (Komoot) geocoding services encountered HTTP 429 (rate limiting) errors and HTTP 403 (forbidden) responses, ultimately resulting in IP-level blocking after sustained usage.

This limitation is particularly problematic for heritage documentation projects in the Global South, where internet connectivity may be unreliable, API costs prohibitive, and institutional relationships with commercial API providers nonexistent. The solution adopted—a specialized local gazetteer with offline fuzzy matching—represents a broader architectural principle of the AEMS framework: designing systems that degrade gracefully under infrastructure constraints rather than failing catastrophically.

4. RESEARCH OBJECTIVES

The present work pursues four primary objectives:

Objective 1. Develop an automated pipeline for extracting structured metadata from community-documented heritage photographs hosted on Wikimedia Commons, including descriptions, dates, authors, and licensing information.

Objective 2. Design and implement a pattern-matching algorithm capable of identifying location references within unstructured description text, using a hierarchical regex strategy with filename-based fallback.

Objective 3. Construct a specialized local gazetteer for Offa, Nigeria, encoding known locations (districts, religious sites, educational institutions, health facilities, landmarks, and businesses) with approximate GPS coordinates, and implement a multi-tier fuzzy matching strategy for offline geocoding.

Objective 4. Generate an interactive, self-contained HTML map of heritage assets, suitable for deployment via GitHub Pages and integration with municipal planning systems, Wikidata knowledge graphs, and community QR-code infrastructure.

Secondary objectives include: (i) providing a reproducible reference implementation for similar deployments in other municipalities; (ii) generating quantitative evidence to support funding submissions to the Wikimedia Foundation and related cultural heritage preservation initiatives; and (iii) demonstrating the adaptability of the AEMS framework beyond environmental monitoring to Digital Public Infrastructure applications.

5. TECHNICAL ARCHITECTURE

The AEMS Heritage Pipeline implements a four-phase sequential architecture, each phase building upon the outputs of the previous phase. The architecture is designed for modularity, reproducibility, and graceful degradation under infrastructure constraints.

5.1 Phase 1: Wikimedia Metadata Extraction

The extraction module queries the Wikimedia Commons API using the `extmetadata` property, which returns structured metadata including `ImageDescription` (user-provided captions), `DateTimeOriginal` (camera timestamp), `Artist` (photographer attribution), and `License` (Creative Commons information).

Implementation details: all requests include a descriptive User-Agent header as required by Wikimedia's API policy, formatted as "AEMS-Heritage-Pipeline/2.1 (UNESP Research Project; contact: celio.s.souza@unesp.br) Python/requests". Requests are rate-limited to 0.3 seconds between calls to respect server load. HTML tags are stripped from descriptions using regex, entities are decoded using Python's `html.unescape` function, and whitespace is normalized.

Output schema: the Phase 1 output is a CSV file with columns: `Filename`, `Commons_URL`, `Latitude` (from EXIF, typically null), `Longitude` (from EXIF, typically null), `Date`, `Has_GPS` (boolean), and `Description`.

5.2 Phase 1.5: Location Pattern Extraction

Location references are extracted from description text using a hierarchical regex strategy ordered from most specific to most generic:

Pattern 1 (prepositional): matches constructions such as "at X", "in X", "held at X", "located at X", "near X", capturing multi-word proper nouns.

Pattern 2 (capitalized sequences): matches capitalized multi-word sequences following prepositions, handling cases where descriptions lack explicit prepositional markers.

Pattern 3 (generic initial): captures the first comma-separated segment of the description, handling descriptions that begin directly with location names.

Pattern 4 (filename fallback): when description-based extraction fails, the algorithm parses the filename itself, removing numeric suffixes (e.g., "01.jpg"), file extensions, and common prefixes (`IMG_`, `DSC_`, `Photo_`), and extracting the first comma-separated segment.

Cleaning pipeline: Extracted location strings undergo a multi-step cleaning process: removal of trailing clauses introduced by "on [date]", "where", "during", or "is a"; removal of leading articles ("The", "A", "An"); whitespace normalization; and minimum length filtering (strings shorter than 4 characters are discarded).

5.3 Phase 2.1: Gazetteer-Based Geocoding

5.3.1 Rationale for Offline Geocoding

Initial implementation attempts using external geocoding APIs (Nominatim, Photon) encountered systematic failures: HTTP 429 rate limiting errors from Nominatim even with 1.1-second delays between requests, and HTTP 403 forbidden responses from Photon, ultimately resulting in IP-level blocking. These failures motivated the design of an offline geocoding module grounded in a specialized local gazetteer.

This architectural decision reflects a broader principle of the AEMS framework: systems designed for data-scarce and infrastructure-constrained environments must minimize external dependencies and degrade gracefully under adverse conditions.

5.3.2 Gazetteer Construction

A specialized gazetteer was constructed with 53 known locations in Offa, organized into the following categories:

- City-level references (4 entries): Offa and variants including state and country qualifiers
- Districts and neighborhoods (19 entries): Ibolo, Ajegunle, Alubata, Balogun, Ojomu, Igbodun, Asalofa, Essa, Shawo, Ilemona, Bolorunduro, Oke-Igbala, Oke-Iyanu, Nawairu deen, Igbo-oro, Ayaba, Anilelerin, Pandora, Ajelanwa
- Religious sites (10 entries): Central Mosque variants, Ajegunle Central Mosque, Akinale Compound Mosque, Anwar-ul Islam Central Mosque, Alubata Oju-irin Mosque, Christ Apostolic Church, ECWA Evangelical Church
- Educational institutions (9 entries): LGEA School variants, ECWA LGEA Primary School, Community LGEA Primary School, Ayo ni o Nursery and Primary School, Bishop Smith LGEA School, Al-hisan School, Ansar-ud-deen College, Anwar-ul Islam Nursery and Primary School, Arolus College of Education
- Health facilities (3 entries): Abike Memorial Clinic, Ajegunle Basic Health Center, Basic Health Care Center
- Landmarks and businesses (8 entries): ONE Innovation Hub, Summit University, Archers Club Hall, Hengee Resort, Ayaba River, A and T Store, Cravepot, De-Unique Kitchen, Ascent Lodge and Club

Each gazetteer entry includes: normalized name (lowercase, stripped), latitude and longitude coordinates (approximate, based on local knowledge and available satellite imagery), and location type classification.

Coordinate sourcing: coordinates were derived from a combination of OpenStreetMap data (where available), satellite imagery analysis, and local knowledge provided by the Offa Heritage Documentation Initiative. For city-level

and district-level entries, coordinates represent approximate centroids; for specific landmarks, coordinates represent approximate building locations. Ground-truthing with GPS devices is planned for future iterations.

5.3.3 Four-Tier Matching Strategy

The geocoding module implements a four-tier matching strategy, ordered from highest to lowest precision:

Tier 1 — Exact match. The normalized input string is compared against normalized gazetteer keys. A match yields the highest precision level ("exact_match").

Tier 2 — Partial match. The algorithm checks whether the normalized input is a substring of any gazetteer key, or vice versa. This handles cases such as "Central Mosque, Offa" matching "central mosque". Precision level: "partial_match".

Tier 3 — Fuzzy match. The algorithm computes the SequenceMatcher ratio (from Python's difflib library) between the normalized input and each gazetteer key, selecting the highest-scoring match above a configurable threshold (default: 0.60). Precision level: "fuzzy_match_[score]".

Tier 4 — City fallback. If the normalized input contains the substring "offa" but no gazetteer match is found, the algorithm returns the city-level coordinates. Precision level: "city_fallback".

5.4 Phase 4: Interactive Map Generation

The map generator uses the Folium library (Python wrapper for Leaflet.js) to produce a self-contained HTML file with the following features:

- Clustered markers via the MarkerCluster plugin, preventing visual overload in dense areas
- Color-coded precision indicators: green (exact_match), blue (partial_match), orange (fuzzy_match), purple (city_fallback)
- Rich popups displaying filename, inferred location, precision level, truncated description (200 characters), and direct link to Wikimedia Commons
- Embedded legend (fixed position, bottom-left) explaining color codes and displaying total asset count
- Embedded title (fixed position, top-center) identifying the map as "Cultural Heritage of Offa, Nigeria"
- OpenStreetMap base tiles, requiring internet access only for tile loading (the HTML file itself is self-contained)

The generated HTML file can be deployed via GitHub Pages, shared via email, or embedded in municipal planning systems without requiring server-side infrastructure.

6. METHODOLOGY

6.1 Data Source

The input dataset consists of 500 images from the Wikimedia Commons category "Digital_Heritage_Documentation_Initiative", documented by the Offa Heritage Documentation Initiative through community photo walks between 2025 and 2026. The images cover heritage assets across the six districts of Offa: Shawo, Essa, Balogun, Ojomu, Asalofa, and Igbodun.

6.2 Execution Environment

The pipeline was executed on a Windows 11 workstation with the following specifications: Python 3.13, pandas 2.0+, requests 2.31+, folium 0.15+. Total execution time was approximately 8 minutes for the full pipeline (Phases 1 through 4), with Phase 1 (Wikimedia API calls) accounting for approximately 3 minutes and Phase 2.1 (gazetteer matching) executing in under 1 second due to the offline design.

6.3 Validation Strategy

Validation was performed at three levels:

Technical validation: the pattern-matching algorithm was manually inspected against a random sample of 50 descriptions to verify that extracted locations corresponded to genuine location references rather than spurious matches.

Geocoding validation: gazetteer coordinates were cross-referenced against satellite imagery (Sentinel-2, 10m resolution) to verify approximate positional accuracy. For city-level and district-level entries, coordinates were verified against OpenStreetMap administrative boundaries.

Ecological validity: the spatial distribution of geocoded assets was reviewed by the local partner (Salako Lukman Olamilekan) to verify alignment with known administrative and cultural geography of Offa.

7. IMPLEMENTATION AND RESULTS

7.1 Phase 1 Results: Metadata Extraction

The Phase 1 module successfully extracted descriptions from all 500 images (100% success rate). The descriptions ranged from short captions (e.g., "Abike Memorial clinic, Offa, kwara state") to detailed narratives (e.g., "The Ibolo Welfare Initiative Program was held at ONE Innovation Hub, Offa, on 25th May 2026, where the Network Intelligence Tracker was officially launched").

EXIF GPS extraction: of the 500 images, zero (0.0%) contained embedded GPS coordinates in their EXIF metadata. This finding confirms the initial hypothesis that community-documented heritage photographs in the Global South

systematically lack geospatial metadata, motivating the development of the offline geocoding module.

7.2 Phase 1.5 Results: Location Pattern Extraction

The pattern-matching algorithm identified location references in 426 of 500 descriptions (85.2% success rate). The remaining 74 descriptions (14.8%) either lacked location references entirely or contained references too ambiguous for extraction (e.g., generic terms such as "This" or references to the documentation initiative itself rather than specific locations).

Unique locations: The 426 extracted location references collapsed to 46 unique locations after deduplication, indicating significant redundancy in the documentation effort (multiple photographs of the same assets).

7.3 Phase 2.1 Results: Gazetteer-Based Geocoding

Of the 46 unique locations identified, 40 (87.0%) were successfully geocoded against the local gazetteer. The six failures corresponded to locations absent from the gazetteer (e.g., "Digital Heritage Documentation Initiatives, Offa", "Vocational center, Offa", "Ecwa J") or to extraction artifacts (e.g., "This", "Mosque, Offa" where the generic term "Mosque" did not match any specific gazetteer entry).

Precision distribution. Among the 40 successfully geocoded locations:

- Exact match: 2 locations (5.0%)
- Partial match: 36 locations (90.0%)
- Fuzzy match: 2 locations (5.0%)

No locations required the city-level fallback, indicating that the gazetteer coverage was sufficient for the identified locations.

7.4 Phase 4 Results: Map Generation

The map generator produced an interactive HTML file containing 321 markers (one per successfully geocoded image), organized into clusters by geographic proximity. The map is centered at coordinates (8.157212, 4.724779), corresponding to the approximate centroid of Offa, with an initial zoom level of 14.

Precision distribution of markers:

- Blue (partial_match): 306 markers (95.3%)
- Green (exact_match): 13 markers (4.1%)
- Orange (fuzzy_match): 2 markers (0.6%)

The predominance of partial matches reflects the design of the matching algorithm, which prioritizes substring containment over exact string equality, and the structure of the input descriptions, which frequently include qualifiers (e.g., "Central Mosque, Offa") beyond the gazetteer keys.

7.5 Summary Statistics

Metric	Value
Total images processed	500
Descriptions extracted	500 (100.0%)
Location references identified	426 (85.2%)
Unique locations	46
Locations geocoded	40 (87.0% of unique)
Images with inferred GPS	321 (64.2% of total)
Execution time	~8 minutes
External API calls (after Phase 1)	0
Reproducibility	Deterministic

8. DISCUSSION

8.1 Strengths of the Approach

Offline-first design: the elimination of external API dependencies after Phase 1 renders the pipeline deployable in low-connectivity environments, a critical requirement for heritage documentation projects in the Global South. The total execution time of 8 minutes on consumer hardware, with zero network dependency for the geocoding phase, demonstrates the feasibility of deployment on modest infrastructure.

Community-grounded gazetteer: the gazetteer coordinates reflect local knowledge rather than generic geocoding results, ensuring alignment with the cultural and administrative geography of Offa. This approach embodies the AEMS principle of embedding domain-specific knowledge into computational architectures.

Transparent uncertainty: each geocoded point carries a precision label (exact_match, partial_match, fuzzy_match, city_fallback), enabling downstream consumers to filter by confidence level. This explicit representation of uncertainty is a core principle of the AEMS framework, extending to cultural heritage applications the same rigor applied to environmental monitoring.

Reproducibility: the deterministic nature of the pipeline, given identical input data, ensures that results can be independently verified and reproduced. The complete source code, gazetteer, and configuration files are publicly available under the MIT license.

8.2 Limitations

Gazetteer coverage: the current gazetteer encodes only 53 locations, representing a fraction of the heritage assets in Offa. Expansion requires community input and ground-truthing with GPS devices, a resource-intensive process that motivates the proposed Wikimedia Foundation grant for scaling.

Coordinate precision: most gazetteer entries use approximate coordinates derived from satellite imagery analysis and local knowledge, with positional accuracy on the order of tens to hundreds of meters. For applications requiring sub-meter accuracy (e.g., structural monitoring, cadastral integration), ground-truthing with GPS devices is necessary.

Pattern extraction recall: the regex-based location extraction algorithm misses complex descriptions where location references are embedded in non-standard syntactic constructions. Named Entity Recognition (NER) models trained on heritage documentation corpora could improve recall, at the cost of increased computational complexity and training data requirements.

Temporal analysis: the current pipeline processes a static snapshot of the Wikimedia Commons category. Extension to temporal analysis—tracking documentation patterns over time, detecting newly documented assets, and monitoring structural degradation through repeat photography—would significantly enhance the system's utility for heritage preservation.

8.3 Comparison with Alternative Approaches

Commercial geocoding services: services such as Google Maps Geocoding API or Mapbox offer higher accuracy and broader coverage but require API keys, incur per-request costs, and introduce dependencies on proprietary infrastructure. For a community-driven heritage documentation project, these trade-offs are unacceptable.

OpenStreetMap-based geocoding: the Nominatim and Photon APIs, while free, exhibited systematic rate limiting and blocking during implementation, rendering them unreliable for batch processing of 500+ images. The offline gazetteer approach, while less comprehensive, offers deterministic behavior and zero external dependencies.

Manual geocoding: manual tagging of 500 images would require approximately 25-50 hours of labor (assuming 3-6 minutes per image), at a cost of \$250-\$500 at typical data entry wages. The automated pipeline accomplishes the same task in 8 minutes at zero marginal cost, representing a 200x improvement in labor efficiency.

9. SCIENTIFIC CONTRIBUTION

The AEMS Heritage Pipeline contributes to three emerging research areas at the intersection of computer science, digital humanities, and development studies.

9.1 AI for Cultural Heritage

The pipeline demonstrates that automated machine learning pipelines can add substantive value to community-driven heritage documentation efforts, transforming raw photographs into structured, geospatially-enabled knowledge infrastructure. This contribution extends the growing literature on AI applications in cultural heritage (e.g., Champion et al., 2022) by providing a reproducible reference implementation optimized for data-scarce environments.

9.2 Decolonial Data Science

By centering local knowledge (the gazetteer) and open infrastructure (Wikimedia) rather than proprietary Western cultural archives, the pipeline embodies a decolonial approach to Digital Public Infrastructure. This contribution aligns with emerging paradigms in decolonial AI (Mohamed, Png, and Isaac, 2020; Couldry and Mejias, 2019) that emphasize technical architectures designed to return infrastructure to communities rather than extract from them.

9.3 Wikimedia and AI Integration

The pipeline provides a scalable method for enriching open knowledge repositories with structured, geospatial data, addressing a persistent challenge in the Wikimedia movement: the labor-intensive nature of semantic structuring and geospatial tagging. By automating these processes, the pipeline enables community documentation initiatives to scale their impact without proportional increases in manual effort.

9.4 AEMS Framework Adaptability

The successful deployment of the AEMS framework in a cultural heritage context—structurally distinct from the environmental monitoring applications for which it was originally designed—demonstrates the framework's domain-agnostic character. This finding supports the broader hypothesis (Souza, 2026) that the core AEMS principles—adaptivity, uncertainty quantification, and offline-first computation—constitute a general-purpose architecture for intelligence systems operating under data scarcity.

10. LIMITATIONS AND FUTURE WORK

10.1 Short-Term Extensions (6-12 months)

Gazetteer expansion: collaborate with the Offa Heritage Documentation Initiative to expand the gazetteer to 200+ locations, covering all six districts comprehensively. Ground-truth coordinates using GPS devices deployed during community photo walks.

NER-based location extraction: train a Named Entity Recognition model on a corpus of heritage documentation descriptions, improving recall for complex syntactic constructions. Evaluate against the regex-based baseline using standard NER metrics (precision, recall, F1).

Wikidata integration: implement automated creation of Wikidata items for geocoded heritage assets, using the Wikibase API to populate properties including coordinate location (P625), located in administrative entity (P131), and image (P18).

Temporal analysis: extend the pipeline to process the Wikimedia Commons category incrementally, detecting newly documented assets and generating time-series analyses of documentation patterns across districts.

10.2 Medium-Term Extensions (12-24 months)

Structural degradation monitoring: integrate computer vision models (e.g., convolutional neural networks trained on architectural degradation datasets) to detect structural changes in heritage assets across repeat photographs, enabling predictive maintenance planning.

Multi-municipality deployment: adapt the pipeline for deployment in other Nigerian municipalities (e.g., Ile-Ife, Kano, Benin City), generalizing the gazetteer construction process and evaluating cross-municipality transferability.

QR-Linked Dynamic Verification Engine: implement the verification engine described in the associated concept note, monitoring QR code scans and flagging broken links or outdated articles to maintain information integrity.

10.3 Long-Term Vision (24+ months)

Pan-African Heritage Infrastructure Network: establish a federated network of municipal heritage pipelines across Africa, sharing architectural components (gazetteer construction, pattern extraction, map generation) while preserving local ownership of cultural narratives.

Integration with AEMS One Health: explore synergies between the heritage pipeline and the AEMS One Health Early Warning System (Souza and Ibrahim, 2026), particularly around shared infrastructure (schools, health centers) that serves both cultural and epidemiological monitoring functions.

11. REPRODUCIBILITY AND OPEN SCIENCE

11.1 Code Availability

The complete source code, configuration files, and documentation are publicly available under the MIT License at: [GitHub repository URL to be inserted]

The repository includes:

- src/core/extractor.py: Wikimedia API interaction and pattern extraction
- src/core/geocoder.py: Gazetteer-based fuzzy matching
- src/core/utils.py: I/O utilities and cache management
- src/pipeline.py: Main orchestrator

- `src/map_generator.py`: Folium-based map generation
- `config/gazetteer.py`: Local gazetteer specification
- `requirements.txt`: Python dependencies
- `README.md`: User documentation
- `reports/TECHNICAL_REPORT.pdf`: This document

11.2 Data Availability

The input dataset (Wikimedia Commons category "Digital_Heritage_Documentation_Initiative") is publicly accessible via the Wikimedia Commons API. The processed output (`offa_data_with_inferred_gps.csv`) and the generated map (`offa_heritage_map.html`) are available in the repository's `data/` and `output/` directories.

11.3 Reproduction Instructions

To reproduce the results:

Step 1: Clone the repository and install dependencies: `git clone https://github.com/SouzaCelio/AEMS-Heritage-Pilot.git > cd aems-heritage-pipeline > pip install -r requirements.txt`

Step 2: Download the input dataset from Wikimedia Commons (script provided in `scripts/download_input.py`).

Step 3: Execute the pipeline: `python -m src.pipeline --input data/offa_raw_data.csv --output data/offa_data_with_inferred_gps.csv`

Step 4: Generate the interactive map: `python -m src.map_generator --input data/offa_data_with_inferred_gps.csv --output output/offa_heritage_map.html`

11.4 Computational Requirements

The pipeline executes on standard consumer hardware (4+ CPU cores, 8+ GB RAM) without specialized accelerators. Total execution time is approximately 8 minutes, dominated by Phase 1 (Wikimedia API calls). Phase 2.1 (gazetteer matching) executes in under 1 second.

12. STRATEGIC ALIGNMENT WITH FUNDING PRIORITIES

12.1 Wikimedia Foundation

The project aligns with the Wikimedia Foundation's strategic priorities around knowledge equity, community-driven content creation, and technical infrastructure for the Global South. Specifically:

- **Knowledge Equity Fund:** The pipeline provides technical infrastructure that enables communities in the Global South to document, structure, and

globally publish their own cultural heritage, directly addressing knowledge gaps in Wikimedia projects.

- **Alliance Grants:** The project's focus on community-driven documentation, open infrastructure, and scalable technical tools aligns with the Alliance Grants funding criteria.
- **Research and Innovation:** The pipeline's novel integration of AI, geospatial analytics, and Wikimedia infrastructure represents a genuine innovation in open knowledge production.

12.2 UNESCO

The project supports UNESCO's mandate around cultural heritage preservation, particularly the 2003 Convention for the Safeguarding of the Intangible Cultural Heritage and the 2005 Convention on the Protection and Promotion of the Diversity of Cultural Expressions. The pipeline's focus on both tangible heritage (buildings, infrastructure) and intangible heritage (Ijakadi Festival, historical narratives) aligns with UNESCO's integrated approach.

12.3 UN Sustainable Development Goals

The project contributes to multiple SDGs:

- **SDG 11 (Sustainable Cities and Communities):** Through municipal heritage mapping and infrastructure documentation
- **SDG 4 (Quality Education):** Through open access to cultural and historical knowledge
- **SDG 9 (Industry, Innovation, and Infrastructure):** Through development of Digital Public Infrastructure
- **SDG 17 (Partnerships for the Goals):** Through international collaboration between Brazil and Nigeria

12.4 African Union Agenda 2063

The project supports the African Union's Agenda 2063 aspiration around "An Africa whose cultural heritage, values and practices are preserved and promoted, and whose creative industries contribute significantly to the continent's economy," by providing technical infrastructure for community-driven cultural preservation.

13. EXPECTED IMPACT AND SCALABILITY

13.1 Direct Impact (Offa Municipality)

The pipeline will deliver:

- A comprehensive, geospatially-indexed inventory of Offa's cultural heritage assets
- An interactive map accessible to municipal planners, educators, tourists, and community members

- Structured Wikidata items enabling integration with global knowledge systems
- A verifiable, anti-misinformation information loop through QR-code infrastructure

13.2 Scalability to Other Municipalities

The pipeline's modular architecture enables deployment in other Nigerian municipalities with minimal adaptation. The primary adaptation required is gazetteer construction, which can be semi-automated through OpenStreetMap data extraction and community validation. Estimated deployment time for a new municipality: 2-4 weeks, including gazetteer construction and community engagement.

13.3 Economic Impact

By enabling cultural tourism, supporting municipal planning, and preserving cultural heritage for future generations, the pipeline generates economic value that exceeds its development cost by orders of magnitude. Preliminary estimates suggest that comprehensive heritage documentation can increase cultural tourism revenue by 15-30% in municipalities with significant heritage assets (World Bank, 2021).

13.4 Democratic Knowledge Access

The pipeline establishes a bulletproof, verified information loop in public spaces, empowering citizens with instant access to fact-checked historical and public health data through QR-code infrastructure. This contribution strengthens democratic knowledge access and counters misinformation vectors.

14. ACKNOWLEDGMENTS

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APPENDIX A: GAZETTEER SPECIFICATION

The complete gazetteer is encoded in config/gazetteer.py and contains 53 entries organized by category. Each entry includes:

- Normalized name (lowercase, stripped)
- Latitude (decimal degrees, WGS84)
- Longitude (decimal degrees, WGS84)
- Type classification (city, district, mosque, church, school, health, landmark, business, river, university)

Coordinate accuracy varies by entry type:

- City-level entries: approximately 100m accuracy
- District-level entries: approximately 200m accuracy
- Landmark-level entries: approximately 50m accuracy (where ground-truthed)

The gazetteer is designed for community expansion; local partners are encouraged to submit additional entries via pull requests to the GitHub repository.

APPENDIX B: PRECISION LEVEL DEFINITIONS

Precision Level	Definition	Typical Positional Accuracy
exact_match	Input string exactly matches a gazetteer key (after normalization)	50-200m (depends on gazetteer entry)
partial_match	Input string is a substring of a gazetteer key, or vice versa	50-200m
fuzzy_match_[score]	SequenceMatcher ratio between input and gazetteer key exceeds threshold (default 0.60)	100-500m (degrades with score)
city_fallback	Input contains "offa" but no specific match found	~1km (city centroid)
failed	No match found at any tier	N/A
no_location_found	Pattern extraction failed to identify a location reference	N/A

APPENDIX C: SAMPLE DATA SCHEMA

The output CSV (*offa_data_with_inferred_gps.csv*) contains the following columns:

Column	Type	Description
Filename	string	Original filename from Wikimedia Commons
Commons_URL	string	Direct URL to the image on Wikimedia Commons
Latitude	float or null	EXIF GPS latitude (typically null)
Longitude	float or null	EXIF GPS longitude (typically null)
Date	string	Image date (from EXIF or Wikimedia metadata)
Has_GPS	boolean	True if EXIF GPS was present
Description	string	Extracted image description
Inferred_Location	string	Location reference extracted by pattern matching
Inferred_Latitude	float or null	Geocoded latitude from gazetteer
Inferred_Longitude	float or null	Geocoded longitude from gazetteer
Display_Name	string	Full display name from gazetteer
Geocoding_Precision	string	Precision level (see Appendix B)
Has_Inferred_GPS	boolean	True if gazetteer geocoding succeeded

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Contact: celio.s.souza@unesp.br

ORCID: 0009-0000-1565-3803